Technology Stack

Date	September 30, 2026
Team ID	6ab2a8af5ccf576cf9a95a906b
Project Name	Cricket Player Performance Tracking Application : CRICKSTATS
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	**React.js** (with Tailwind CSS & Chart.js)	Offers component reusability and interactive, responsive UI rendering for real-time statistical visualization and player performance analytics.

2	**Backend / Server-Side**	**Node.js** (Express.js)	Provides asynchronous, non-blocking I/O ideal for handling multiple concurrent requests, fast processing of player stats, and scalable RESTful API integration.
3	**Database / Data Storage**	**PostgreSQL** (or MongoDB)	Ensures strong relational mapping for structured cricket data (players, matches, stats, teams) along with fast query performance and ACID compliance.
4	**Cloud / Hosting / Deployment**	**Docker & AWS** (EC2 / S3)	Containerizes the application for consistent environment delivery, easy scaling, dynamic load balancing, and reliable continuous hosting.
5	**Version Control & CI/CD**	GitHub & GitHub Actions	Enables seamless team collaboratio automated deployment pipelines upon code commits.